

Qatar-1b

R-2531

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_210625 · created 2026-08-25T23:09:30+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2459390.81389 +/- 0.00096 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1743 +/- 0.0082
Transit depth (Rp/Rs)^2	3.04 +/- 0.29 [%]
Orbital Inclination (inc)	86.48 +/- 1.03
Airmass coefficient 1 (a1)	1.442 +/- 0.024
Airmass coefficient 2 (a2)	-0.293 +/- 0.022
Scatter in the residuals of the lightcurve fit is	1.63 %
Best Comparison Star	None
Optimal Aperture	2.74
Optimal Annulus	10.93
Transit Duration (day)	0.0802 +/- 0.0032

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1743 ± 0.0082 vs 0.1463 (deviation 2.69σ)
Duration fit vs published	0.0802 d vs 0.0692 d (deviation 2.72σ)
Mid-time O–C	-5.1 min
Depth SNR	16.8
Residual scatter	1.63 %

Light curve

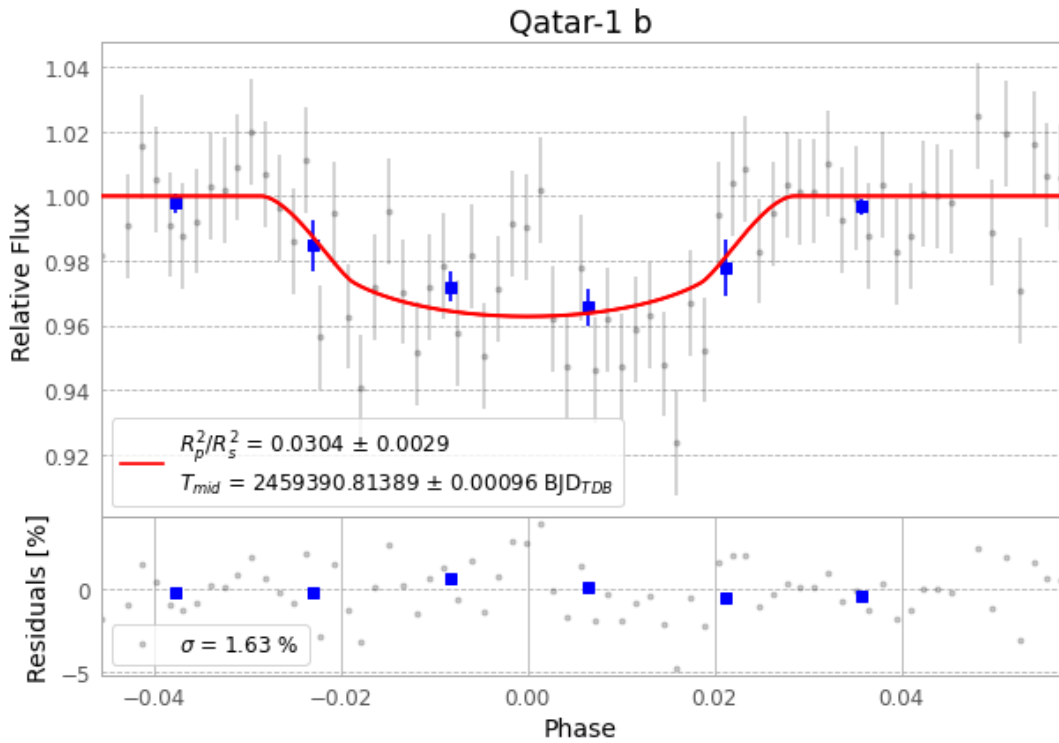


Figure 1 — FinalLightCurve_Qatar-1 b_2021-06-24.png (SHA-256 f5f07178e9556ee8...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [341, 261], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 91b2071177694cc4...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

70 FITS frames (46 MB), Qatar-1210625055715.FITS → Qatar-1210625092712.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-210625.log (1f481cb09226...), FinalLightCurve_Qatar-1 b_2021-06-24.png (f5f07178e955...), FinalLightCurve_Qatar-1 b_2021-06-24.csv (fe75c12d64a6...)

Compute resources

Wall time 117.8 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-25 23:09Z.